

WEEK 7 – POWER BI REPORT

Introduction

The Week 7 activity was carried out using Microsoft Power BI to prepare a Student Performance dataset for further analysis.

The dataset contains information related to students, including gender, race/ethnicity, parental education, lunch type, test preparation, and marks obtained in Mathematics, Reading, and Writing.

The main purpose of this activity was to clean the dataset, transform the available information, create additional performance-related columns, and prepare the data for dashboard development.

Objectives

The main objectives of this activity are:

*To import the student performance dataset into Power BI.

* To examine the structure and contents of the dataset.

*To identify missing, duplicate, and incorrect data.

* To clean and standardize the dataset.

*To change columns to suitable data types.

*To calculate Total Score and Average Score.

*To categorize students according to their performance.

*To create Pass and Fail classifications.

*To convert subject-wise marks into a suitable format using Unpivot.

- *To prepare the dataset for modelling.

- *To create basic DAX measures.

- *To verify the final transformed data.

Dataset

The Student Performance dataset contains academic and background information about students.

Important Columns

- *Gender

- * Race/Ethnicity

- * Parental Education

- * Lunch

- * Test Preparation Course

- * Math Score

- * Reading Score

- * Writing Score

Importing the Dataset

The dataset was first loaded into Microsoft Power BI Desktop.

Steps:

*Power BI Desktop → Get Data → Excel → Select Dataset → Load.

*After selecting the dataset, "Transform Data" was chosen to open the Power Query Editor.

The screenshot displays the Microsoft Power Query Editor interface. The main area shows a table with columns: Student ID, gender, race/ethnicity, parental level of education, lunch, and test preparation course. Each column has a data type dropdown and a status indicator (Valid, Error, Empty) with a percentage. The table contains 22 rows of data. On the right, the 'Query Settings' pane is open, showing the 'Name' as 'StudentsPerformance'. Below it, the 'APPLIED STEPS' list includes: Source, Promoted Headers, Changed Type, Capitalized Each Word, Changed Type1, Cleaned Text, Trimmed Text, Added Index, Renamed Columns, Reordered Columns, Added Custom, Added Custom1, Changed Type2, and 'Added Conditional Column' (which is currently selected).

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed

Understanding the Data

Before making any changes, the dataset was examined in Power Query Editor.

The following points were checked:

- *Number of rows and columns
- *Column names
- * Data types
- * Numerical columns
- * Categorical columns
- * Blank or missing values
- * Error values
- * Duplicate records

This inspection helped to understand the dataset before applying transformations.

Data Cleaning

The dataset was cleaned to improve its quality and consistency.

The following operations were performed:

- * Removed unnecessary duplicate records.
- * Checked and handled blank values.
- * Identified invalid values.
- * Corrected unsuitable data types.
- * Standardized categorical information.
- * Renamed columns wherever necessary.

These operations helped to make the dataset more suitable for analysis.

The screenshot displays a data transformation tool interface. The main table contains student data with columns: Student ID, gender, race/ethnicity, parental level of education, lunch, and test preparation course. Each column has a data quality summary row showing Valid, Error, and Empty percentages. The 'test preparation course' column is highlighted in blue. To the right, the 'Query Settings' panel shows the 'Name' as 'StudentsPerformance' and a list of 'APPLIED STEPS' including Source, Promoted Headers, Changed Type, Capitalized Each Word, Changed Type1, Cleaned Text, Trimmed Text, Added Index, Renamed Columns, Reordered Columns, Added Custom, Added Custom1, Changed Type2, and Added Conditional Column.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed
23	Male	Group D	Some College	Standard	None

Categorical Data Transformation

The categorical columns were reviewed to ensure that the values were consistent.

The main categorical fields included:

- *Gender
- *Race/Ethnicity
- *Parental Education
- *Lunch
- *Test Preparation Course

Standardizing these values makes filtering and comparison easier when creating Power BI visualizations.

Creating Total Score

*A new calculated column named "Total Score" was created.

*The total marks were calculated using the three subject scores:

****Total Score = Math Score + Reading Score + Writing Score.***

*This column provides the combined score of each student.

Creating Average Score

*Another calculated column named "Average Score" was created.

The calculation used was:

*Average Score = Total Score / 3.

The Average Score provides an overall indication of a student's performance across the three subjects.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Grade 

	Column Name	Operator	Value		Output
If	Average	is greater than or...	80	Then	A
Else If	Average	is greater than or...	60	Then	B
Else If	Average	is greater than or...	40	Then	C

Add Clause

Else

Fail

OK

Cancel

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Result



	Column Name	Operator	Value		Output	
If	Average	is greater than or...	ABC 123 40	Then	ABC 123 Pass	...

Add Clause

Else

ABC 123

Fail

OK

Cancel

Performance Classification.

Students were grouped into different performance levels using conditional logic.

Grade Performance Level

A High Performance

B Good Performance

C Average Performance

F Low Performance

This classification helps to identify students based on their overall academic results.

Pass and Fail.

A conditional column was created to classify students based on the selected pass criteria.

The resulting categories were:

*Pass.

*Fail.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Result 

	Column Name	Operator	Value		Output	
If	Average	is greater than or...	ABC 123 40	Then	ABC 123 Pass	...

Add Clause

Else

ABC 123 Fail

OK

Cancel

Unpivoting Subject Scores.

The three subject columns were selected:

*Math Score.

*Reading Score.

*Writing Score.

The Unpivot Columns option was used in Power Query.

After transformation, the columns were renamed as:

***Subject.**

***Score.**

Queries [4]

StudentsPerformance

Student Marks

Student Details

Measure

Student ID Subject Score

Valid 100% Valid 100% Valid 100%

Error 0% Error 0% Error 0%

Empty 0% Empty 0% Empty 0%

1	1	math score	72
2	1	reading score	72
3	1	writing score	74
4	2	math score	69
5	2	reading score	90
6	2	writing score	88
7	3	math score	90
8	3	reading score	95
9	3	writing score	93
10	4	math score	47
11	4	reading score	57
12	4	writing score	44
13	5	math score	76
14	5	reading score	78
15	5	writing score	75
16	6	math score	71
17	6	reading score	83
18	6	writing score	78
19	7	math score	88

Data Model Preparation

After completing the transformations, the dataset was prepared for the Power BI data model.

The following checks were performed:

- * Reviewed the transformed table.
- * Removed columns that were not required.
- * Checked the required keys.
- * Reviewed possible relationships.
- * Prepared the table for visualization.

The purpose of this step was to ensure that the data was organized properly before loading it into the report.

Loading the Transformed Data

*After completing all required transformations, Close & Apply was selected.

*The cleaned dataset was then loaded back into Power BI Desktop.

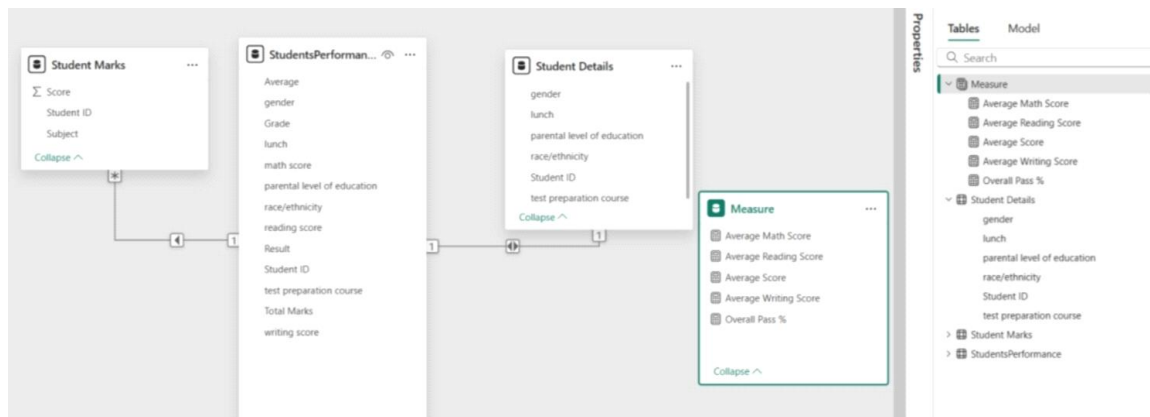
*The calculated columns and transformed subject data were checked after loading.

DAX Measures.

Basic DAX measures were created for analysing student performance.

The measures include:

- * Average Math Score.
- * Average Reading Score.
- * Average Writing Score.
- * Overall Average Score.
- * Overall Pass %.



Data Validation.

After completing the transformation process, the final dataset was checked for accuracy.

The following were verified:

- * Calculated Total Score.
- * Average Score.
- * Pass/Fail classification.
- * Performance Grade.
- * Transformed subject data.
- * Overall data structure.

The Power Query results were also compared with the original dataset to ensure that the transformation process was completed correctly.

Final Output.

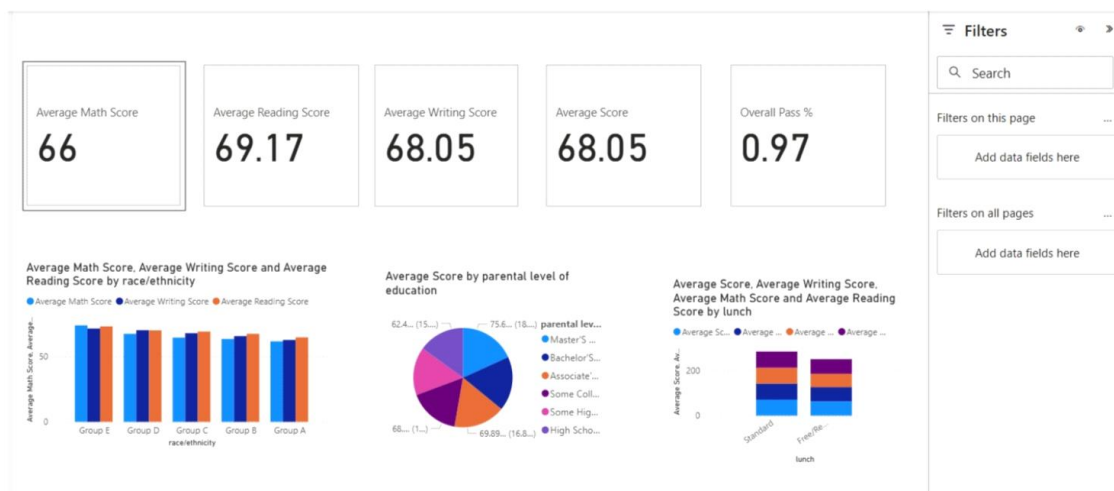
The final output of the Week 7 activity is a cleaned and transformed Student Performance dataset.

The prepared dataset contains:

- * Cleaned student information
- * Total Score

- * Average Score
- * Performance Grade
- * Pass/Fail classification
- * Subject and Score columns
- * Basic DAX measures

This prepared data can be used for creating the Student Performance Analytics Dashboard in the next week.



Conclusion

The Week 7 Power BI activity helped in understanding the complete process of preparing a dataset for analysis.

The Student Performance dataset was imported, inspected, cleaned, transformed, and validated using Power Query Editor. Additional performance columns and classifications were created, and the subject-wise scores were unpivoted.

The final dataset is now organized and ready for developing an interactive Power BI dashboard with KPIs, charts, filters, and slicers.

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